Can I trust an automated equity research report?

FinRobot audit and a data-driven forecast improvement

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Finding. The saved reports are useful drafts but are not ready for portfolio decisions. A 20-item filing audit finds 16 matches and four explained EBITDA-definition differences. Of 288 manually extracted statements across three companies and eight sections, only 88 (30.6%) are supported by their supplied evidence. Fifty (17.4%) are contradicted. The report presentation also converts missing target prices to zero and replaces valid competitor prose with a generic fallback.

Measured improvement. A bounded historical-median forecast lowers revenue mean absolute percentage error from 14.0% to 10.2% and EBITDA weighted absolute error from 29.5% to 20.1% on ten retrospective company-year forecasts. Revenue median error worsens and Intel remains difficult. These results support a narrower numerical improvement, not an improvement in generated prose or investment performance.

Coverage and execution

Ticker	Peers	Last actual FY	Compute seconds	LLM cost (USD)
NVDA	AMD, INTC	2026A	29.63	0.03926
AMD	NVDA, INTC	2025A	30.97	0.03975
INTC	AMD, NVDA	2025A	32.77	0.03973
AAPL	CSCO, NVDA	2025A	28.18	0.04019
CSCO	AAPL, NVDA	2026A	26.86	0.03927

All five captured FMP profiles classify the companies as Technology. NVIDIA, AMD and Intel provide contrasting growth histories; Apple and Cisco extend coverage to technology hardware. This is a broad sector universe: consumer devices and networking have different economics from chip firms. Peer comparisons are descriptive and should not be treated as interchangeable valuation benchmarks. An initial QCOM attempt returned HTTP 402 on all five statement/ratio endpoints and made no model calls; AVGO was not run. These names are excluded from successful coverage.

The baseline uses Python 3.10.6, OpenAI 1.109.1, model alias gpt-4.1, actual snapshot gpt-4.1-2025-04-14, temperature 0.7 and max_tokens 1000. The FinRobot commit is 2717499b8e30f242640af08c4ad9afd1113c2d45, with a separately archived FMP Stable compatibility patch. Analysis and rendering times are added; hours spent waiting between stages are excluded.

49 unique captured completions used 70,714 input tokens (7,680 cached) and 16,835 output tokens. Estimated recorded spend is $0.264588, using $2/$0.50/$8 per million uncached input/cached input/output tokens [4]. Unrecorded setup checks are excluded; this is not a billing export. FMP incremental cost is assumed zero on the existing Basic plan.

1. Filing reconciliation

Ten metrics in the master CSV were checked for each of NVIDIA and Apple against FY2025 SEC 10-K inline XBRL facts [1,2]. NVIDIA FY2025 ends 26 January 2025; Apple FY2025 ends 27 September 2025. The audit matches fiscal periods rather than calendar-year labels. Money below is USD billions except basic EPS; margins and revenue growth are percent. M = match within rounding; D = explained definition difference.

NVDA - FY2025

Metric	Pipeline	Filing comparable	Delta	Verdict
Revenue	130.497	130.497	+0.000	M
Cost of Operations	32.639	32.639	+0.000	M
SG&A	3.491	3.491	+0.000	M
Contribution Profit	97.858	97.858	+0.000	M
Contribution Margin	75.00	74.99	+0.01	M
EBITDA	86.137	83.317	+2.820	D
EBITDA Margin	66.00	63.85	+2.15	D
SG&A Margin	2.70	2.68	+0.02	M
Revenue Growth	114.20	114.20	-0.00	M
EPS	2.97	2.97	+0.00	M

AAPL - FY2025

Metric	Pipeline	Filing comparable	Delta	Verdict
Revenue	416.161	416.161	+0.000	M
Cost of Operations	220.960	220.960	+0.000	M
SG&A	27.601	27.601	+0.000	M
Contribution Profit	195.201	195.201	+0.000	M
Contribution Margin	46.90	46.91	-0.01	M
EBITDA	144.427	144.748	-0.321	D
EBITDA Margin	34.70	34.78	-0.08	D
SG&A Margin	6.60	6.63	-0.03	M
Revenue Growth	6.40	6.43	-0.03	M
EPS	7.49	7.49	+0.00	M

The filing comparable EBITDA is operating income plus cash-flow D&A. NVIDIA gives 81.453 + 1.864 = 83.317B. FMP gives pretax income 84.026 + interest 0.247 + D&A 1.864 = 86.137B. The 2.820B gap is net other income 2.573B plus interest 0.247B; D&A agrees. Apple operating EBITDA is 133.050 + 11.698 = 144.748B, versus FMP pretax income plus D&A of 144.427B, a -0.321B net nonoperating gap.

EBITDA is a derived, non-GAAP measure here, not a reported filing line. Contribution Profit maps to gross profit; EPS uses basic EPS. Money tolerance is $1, EPS $0.005 and percentage tolerance 0.05 percentage points. Exact tags, contexts, URLs and explanations are in data_audit.csv.

2. Forecast and valuation assumptions

Component	Baseline	Historical-median method
Revenue	Same 5%, 6%, 4% forward rates for every firm.	Recent median annual growth, bounded and gradually returned toward a 5% anchor.
Gross margin / SG&A	+1 pp gross margin; -0.5 pp SG&A each year.	Trailing three-year medians of dollar amounts divided by revenue; no annual uplift.
EBITDA	Actual FMP EBITDA; future gross profit minus SG&A.	Trailing FMP EBITDA/revenue median, keeping the actual definition.
Forecast dates	Fixed 2025E-2027E; seven labels overlap actual years.	Latest actual fiscal year +1, +2, +3; zero overlaps.
EPS / P/E	EPS scales with revenue; P/E falls 5% annually.	EPS scales with revenue; fixed last-actual implied price; missing P/E for losses.
CAGR	Historical CAGR has an ambiguous heading.	Explicit Historical CAGR label.

Baseline EBITDA is recomputed after the configured +1 pp EBITDA-margin assumption and overwrites it. Gross profit minus SG&A excludes other expenses such as R&D, creating a definition break from actual FMP EBITDA. NVIDIA also has $183.7B in the main displayed 2027E forecast versus $382.0B in the peer 2027 projection. Both enter the prompt. The improvement changes the master forecast only; it does not synchronize peer forecasts.

Independent EV/EBITDA calculation

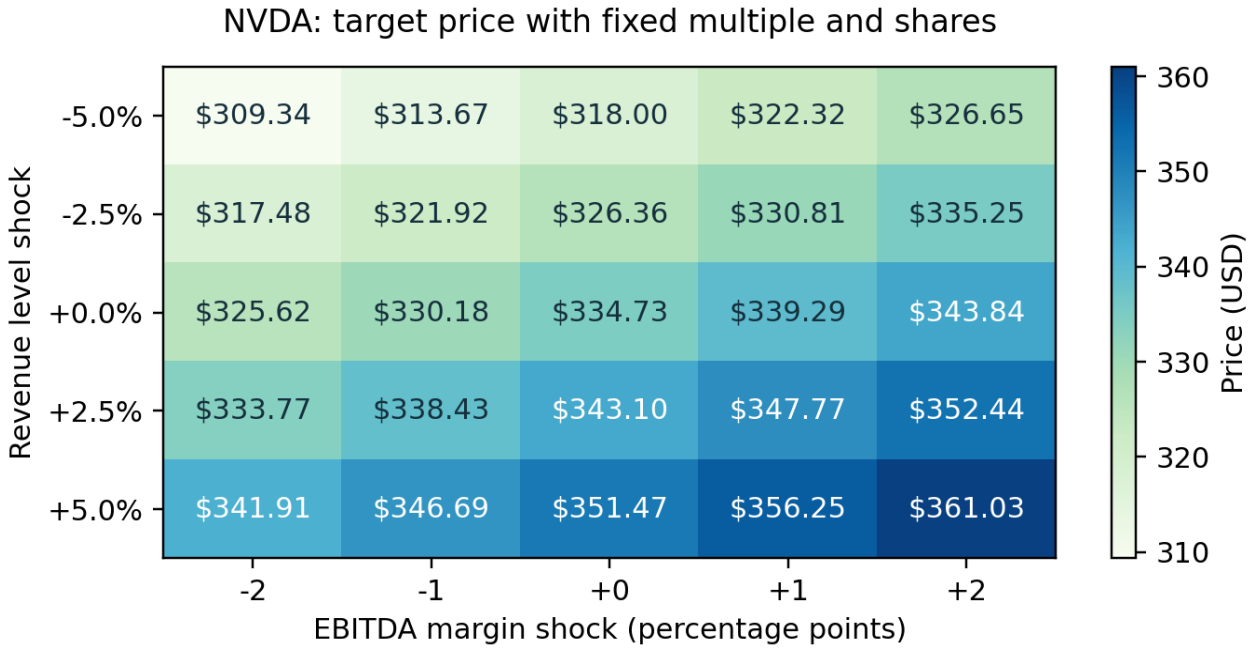
For all five firms, the notebook computes **price = 0.90 x final-forecast EBITDA x mean historical EV/EBITDA / shares**. Shares are current quote market cap divided by the price in that same captured quote. The engine is then invoked separately with those same inputs. Maximum absolute reconciliation error is below $0.00000001.

Ticker	Mean multiple	Hand / engine price	Using balance net debt
NVDA	48.88x	$334.73	$371.89
AMD	40.65x	$377.84	$420.47
INTC	31.52x	$105.28	$110.58
AAPL	23.00x	$304.20	$332.80
CSCO	16.03x	$124.46	$132.62

Hidden assumptions: final E-column EBITDA; arithmetic mean of five historical multiples; net debt = 10% of EV; no discounting, WACC or terminal growth in this method. Its low/high bounds use mean +/- sample standard deviation while holding base proxy debt fixed. The fallback 12x/3x values are unused. The effective third horizon is FY2028 or FY2029, despite the 2027E header. These are undiscounted forward values, not present-value recommendations. The balance-debt alternative uses FMP cash-only net debt and illustrates convention sensitivity.

3. Price sensitivity

The saved pipeline already ran sensitivity_analyzer.py for each company. Its grids report revenue or EBITDA, not a target price. I connect the same revenue-level shocks of +/-5% and EBITDA-margin shocks of +/-2 percentage points to the independently reproduced EV/EBITDA formula. The multiple, shares and 10%-of-EV debt rule stay fixed.



Ticker	Base price	Low corner	High corner	Change range
NVDA	$334.73	$309.34	$361.03	-7.6% to +7.9%
AMD	$377.84	$341.85	$415.62	-9.5% to +10.0%
INTC	$105.28	$93.48	$117.77	-11.2% to +11.9%
AAPL	$304.20	$276.09	$333.67	-9.2% to +9.7%
CSCO	$124.46	$113.14	$136.31	-9.1% to +9.5%

A revenue-only shock changes price by exactly +/-5% under this formula. Combined corners move price roughly -7.6%/+7.9% for NVIDIA and -11.2%/+11.9% for Intel, whose final forecast EBITDA margin is lower. This is a local arithmetic result; it does not validate the forecast, multiple or debt assumptions. Intel 2024 EV/EBITDA of 106.9x can strongly affect a five-year historical average.

The price grid uses the full-precision EBITDA/revenue ratio. The original analyzer starts from a one-decimal margin and rounds its combined grid to $0.1B. Using those displayed values would introduce rounding drift in the base price. This is a revenue-level shock, not an annual growth-rate change compounded for three years, and the range is not a confidence interval.

4. Grounding: evidence and prose

I manually atomized factual, numerical and interpretive statements from the eight saved sections for NVIDIA, AMD and Intel. Named product lists and numerical time series count as one proposition when they express one relationship. Separately checkable judgments and numerical comparisons are split. Boilerplate and titles are excluded. Every substantive paragraph is represented, with exact excerpts and evidence pointers in grounding_audit.csv.

Rubric. Supported means directly derivable from that company's captured prompt; Contradicted means the prompt says otherwise, including actual/forecast or year confusion. Unsupported means absent evidence, including prior knowledge, causal interpretations and conditional risk scenarios. Unsupported is not synonymous with false. Financial rounding is allowed; the coarse $26.9B NVIDIA figure is within 0.5% of $26.974B.

Section	Supported	Unsupported	Contradicted	Total	Rate
Company Overview	12	45	8	65	18.5%
Competitor Analysis	17	26	7	50	34.0%
Investment Overview	22	20	10	52	42.3%
Major Takeaways	0	12	0	12	0.0%
News Summary	0	3	0	3	0.0%
Risks	10	18	6	34	29.4%
Tagline	6	9	3	18	33.3%
Valuation Overview	21	17	16	54	38.9%
All sections	88	150	50	288	30.6%

Structural reasons

The audited CLI imports modules/text_generator_agents.py and makes sequential plain Chat Completions. It does not call the separate typed openai-agents/Pydantic manager described in the assignment. Its active risks prompt asks for five concise bullets, not the unused 600-800-word SDK risk-agent prompt. Seven model calls produce eight saved sections because news is unavailable.

The shared user evidence contains the master financial table and two peer tables. Company-overview instructions ask for products, business model and market position without supplying those facts. Valuation instructions ask for current metrics and fair value without supplying a current quote or intrinsic valuation. An ambiguous CAGR heading and overlapping A/E dates invite confident temporal errors. The company-overview section has the lowest model-generated support rate, 18.5%; valuation has 16 contradicted claims out of 54, the largest contradicted count.

Examples: AMD 38.3x is called higher than Intel 106.9x; NVIDIA historical 68.3% CAGR becomes a forward growth rate. A $382B NVIDIA peer forecast is prompt-supported but conflicts with the master forecast. The twelve generic takeaway claims are postprocessor fallbacks, and three no-news claims are API-unavailable placeholders. Their zero support rates describe the saved pipeline output, not separate model hallucination rates.

5. Same-input stability and rendering

NVIDIA was run three times with byte-identical financial CSVs and exactly identical JSON model request bodies, including prompts, temperature and max_tokens. Repeat runs replay the archived FMP responses but make fourteen new, successful model calls. The returned model snapshot is unchanged. No FMP network call occurs in the repeat study.

Run	Headline rating	Valid target	Canonical top-three risks
1	Overweight	Missing	Valuation; Growth; Competition
2	Overweight	Missing	Valuation; Growth; Margin
3	Overweight	Missing	Valuation; Growth; Margin

Run pair	Top-three set Jaccard	Same-rank agreement
1 / 2	0.50	2 / 3
1 / 3	0.50	2 / 3
2 / 3	1.00	1.00
Mean	0.667	0.778

Risk headings were normalized into semantic categories before comparison. Competition appears in the top three once and Margin twice; both presence indicators have sample variance 1/3. Valuation and Growth occur three times and have variance zero. This three-run sample demonstrates wording/priority variation, not a well-estimated long-run distribution.

Headline rating comes from frozen FMP analyst grades, not a model recommendation. Its ordinal sample variance is zero, a control check. The displayed target price is 0.0 after price-target-news returns HTTP 402 in all three runs. There are zero valid price observations, so target-price variance is undefined, not zero. Stable headlines do not establish stable LLM investment conclusions.

Deterministic presentation failures

All five baseline competitor texts contain more than 20 commas. The renderer interprets them as CSV data and replaces them with a generic competitor paragraph when text regeneration is disabled. All five combined HTML files therefore omit the original competitor prose even though every analysis-file hash stays unchanged. Auditing the HTML alone would miss the original claim errors.

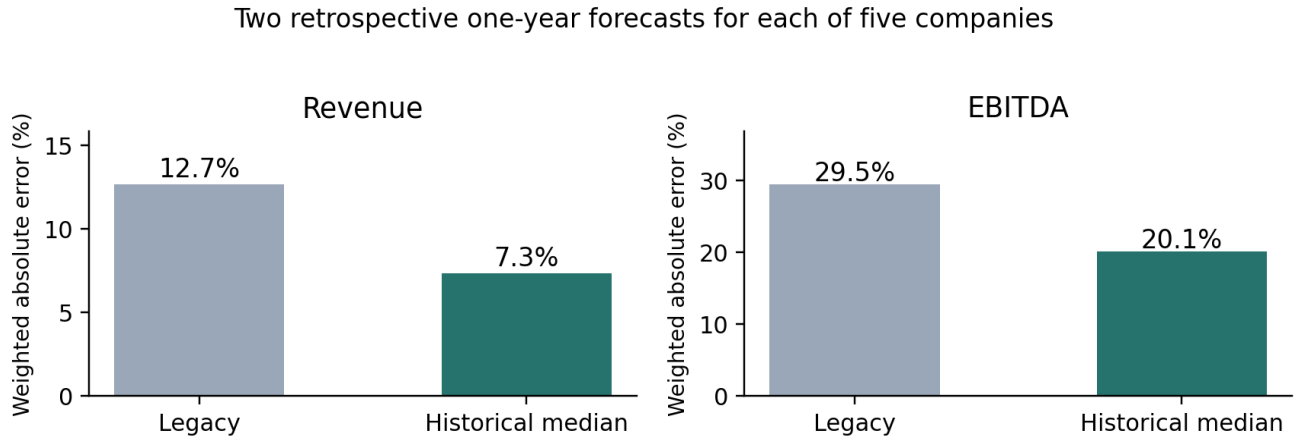
All five major-takeaway model responses are successfully captured, but the exact "Revenue Growth:" guard rejects their formatting and saves generic text. Blocked news retrieval creates "No recent news available," which does not establish absence of events. Source acquisition, interpretation and rendering all need separate validity checks.

Evidence: stability_audit.csv, stability_pairs.csv, stability_metrics.json, coverage.csv, and the archived stability manifest. Request/response bodies and source snapshots allow a reader to distinguish live completions from replayed financial data.

6. Before and after: forecast errors

I chose Option B and evaluated the same five companies in two expanding-window folds: train on the earliest three annual observations to predict the fourth, then on four to predict the fifth. Each is a one-year-ahead forecast. Legacy constants remain 5%, 6%, 4%, but estimate labels are moved to cutoff +1/+2/+3 for the retrospective comparison. Calibration never reads the target-year row.

Metric / statistic	Legacy	Historical median	Better folds
Revenue mean APE	14.03%	10.21%	6 / 10
Revenue median APE	6.64%	7.57%	-
Revenue WAPE	12.69%	7.33%	-
EBITDA mean APE	168.66%	134.31%	8 / 10
EBITDA median APE	39.10%	19.37%	-
EBITDA WAPE	29.49%	20.14%	-



Company	Revenue MAPE: before / after	EBITDA WAPE: before / after
NVDA	43.8% / 19.7%	36.8% / 33.7%
AMD	14.7% / 10.4%	57.3% / 19.3%
INTC	6.4% / 14.5%	113.1% / 121.3%
AAPL	2.1% / 2.3%	16.6% / 4.4%
CSCO	3.2% / 4.2%	37.0% / 11.7%

Revenue improves mainly for NVIDIA and AMD; Intel worsens in both folds. NVIDIA EBITDA FY2025 and Intel EBITDA FY2025 also worsen. Intel FY2024 EBITDA is only $1.203B, so percentage errors exceed 1,000% and dominate EBITDA MAPE. WAPE weights absolute dollar errors by aggregate actual dollars; it is less sensitive to that denominator but gives larger firms more weight. Neither metric alone establishes forecast quality.

7. Design, limits and decision standard

The improved first growth rate is the median of the latest up-to-three annual revenue changes, clipped to [-25%, +50%]. With only three annual training observations, two changes are available. Later rates are **5% + (first growth - 5%) x 0.5^(horizon - 1)**. Gross, SG&A and FMP EBITDA margins use medians of the latest three annual dollar ratios. First production growth assumptions are NVIDIA 50.0%, AMD 13.7%, Intel -2.1%, Apple 2.0% and Cisco 5.3%.

The bounds, 5% anchor and half-decay are explicit judgmental guardrails, not fitted economic parameters. They limit extrapolation and retain cross-company variation. EPS assumes unchanged latest net margin and shares; P/E holds the last-actual implied price fixed and is missing for losses. Three consecutive annual periods and valid positive revenues are required. The CLI rejects quarterly mode before API retrieval.

The clean improvement patch touches only financial_data_processor.py and the CLI. Default legacy behavior is unchanged. Five-company legacy outputs were checked against the original processor, actual-history columns were preserved exactly, and eight boundary tests pass. A separate infrastructure patch records the Stable API migration used by the baseline; it is not counted as the forecast improvement.

Limits

Ten folds are a small, retrospective sample. FMP data were downloaded in one current vintage and may contain restatements or split adjustments; this is not a point-in-time backtest. No confidence or significance claim is made. Hypergrowth caps can understate NVIDIA, historical decline can overstate Intel weakness, and medians miss restructuring and product-cycle reversals. The experiment tests one-year errors, not the three-year path. It uses the FMP EBITDA definition, not an economically normalized EBITDA measure.

No additional LLM report was generated for the improved forecasts. The measured after-results concern numerical forecasts only. Peer forecast discrepancies, missing news, missing headline targets and presentation fallbacks remain. A single reviewer annotated grounding; claim granularity and conservative treatment of judgments affect the rates. Raw excerpts are supplied for independent relabeling.

Would I send this to a portfolio manager?

I would share the reconciled historical revenue, cost, SG&A, gross-profit and basic-EPS rows with their fiscal periods and sources. EBITDA needs a definition label. I would not present the narrative fair-value conclusion, zero target, business background, generic takeaways or current peer multiple discussion as decision-grade research. Before use, the pipeline needs synchronized fiscal horizons, consistent EBITDA, explicit debt/share/multiple/discount assumptions, business and news evidence, citation-aware verification, and checks that rendered text matches approved source text.

Sources and reproducibility

[1] NVIDIA FY2025 Form 10-K, SEC EDGAR. [2] Apple FY2025 Form 10-K, SEC EDGAR.
[3] FinRobot source at the pinned commit. [4] OpenAI GPT-4.1 model and token pricing.
[5] Course assignment requirements. Retrieved 12 October 2026. README.md gives commands; audits/, improvement/, valuation_reproduction.ipynb and artifacts/ contain calculations and original evidence.